

HDOC-27

Question:

Write a C program that displays the following menu and uses a switch statement to print the “Pyramid/Diamond pattern” selected by the user. Use nested loops without arrays. Each pattern may be implemented in a separate parameterless function with no return value.

Task1: Algorithm:

Main Program

1. Start.
2. Declare an integer variable choice.
3. Display the menu:
 - Full Pyramid
 - Inverted Pyramid
 - Diamond Pattern
 - Palindromic Number Pyramid
 - Repeated-Number Pyramid
 - Exit
4. Read choice.
5. Use a switch statement:
 - If choice = 1, call fullPyramid().
 - If choice = 2, call invertedPyramid().
 - If choice = 3, call diamond().
 - If choice = 4, call palindromicPyramid().
 - If choice = 5, call repeatedPyramid().
 - If choice = 6, exit.
 - Otherwise, display "Invalid choice!".
6. Repeat the menu until the user selects Exit.
7. Stop.

Algorithm for Full Pyramid

1. Input number of rows n.
2. Repeat for i = 1 to n.
3. Print n-i spaces.
4. Print 2*i-1 asterisks.
5. Move to the next line.

This produces the required odd-number sequence of stars: 1, 3, 5, 7, etc.

Algorithm for Inverted Pyramid

1. Input n.

2. Repeat from $i = n$ down to 1.
3. Print $n-i$ spaces.
4. Print $2*i-1$ asterisks.
5. Move to the next line.

The assignment requires the inverted pyramid to begin with $2n-1$ stars and decrease toward one star.

Algorithm for Diamond Pattern

1. Input n .
2. For the upper half, repeat $i = 1$ to n .
3. Print $n-i$ spaces and $2*i-1$ stars.
4. For the lower half, repeat $i = n-1$ down to 1.
5. Print $n-i$ spaces and $2*i-1$ stars.
6. Stop after printing a total of $2n-1$ rows.

This follows the specified diamond structure.

Algorithm for Palindromic Number Pyramid

1. Input n .
2. Repeat $i = 1$ to n .
3. Print suitable leading spaces.
4. Print numbers from 1 to i .
5. Print numbers from $i-1$ down to 1.
6. Move to the next line.

For example, row 4 becomes 1 2 3 4 3 2 1.

Algorithm for Repeated-Number Pyramid

1. Input n .
2. Repeat $i = 1$ to n .
3. Print suitable leading spaces.
4. Print the value i , $2*i-1$ times.
5. Move to the next line.

Thus, row 4 contains seven 4s.

Task2: Test case table:

Test Case	Menu Choice	Input n	Test Type	Expected Result
1	1	4	Valid	Full pyramid with 4 rows
2	2	4	Valid	Inverted pyramid with 4 rows
3	3	4	Valid	Diamond containing 7 rows
4	4	4	Valid	Palindromic pyramid with 4 rows
5	5	4	Valid	Repeated-number pyramid with 4 rows
6	1	1	Boundary	Prints one *
7	3	1	Boundary	Diamond consists of one *
8	4	1	Boundary	Prints 1
9	5	1	Boundary	Prints 1
10	7	—	Invalid	Displays Invalid choice!
11	6	—	Exit	Displays Exiting program... and terminates

Task3: C program:

```
#include <stdio.h>
void fullPyramid();
void invertedPyramid();
void diamond();
void palindromicPyramid();
void repeatedPyramid();
int main()
{
    int choice;
    do
    {
        printf("\n--- PYRAMID / DIAMOND PATTERNS ---\n");
        printf("1. Full Pyramid\n");
        printf("2. Inverted Pyramid\n");
        printf("3. Diamond Pattern\n");
        printf("4. Palindromic Number Pyramid\n");
        printf("5. Repeated-Number Pyramid\n");
        printf("6. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice)
        {
            case 1:
                fullPyramid();
                break;
            case 2:
                invertedPyramid();
                break;
```

```

        case 3:
            diamond();
            break;
        case 4:
            palindromicPyramid();
            break;
        case 5:
            repeatedPyramid();
            break;
        case 6:
            printf("Exiting program...\n");
            break;
        default:
            printf("Invalid choice!\n");
    }
} while(choice != 6);
return 0;
}
/* Pattern 1: Full Pyramid */
void fullPyramid()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n - i; j++)
        {
            printf(" ");
        }
        for(j = 1; j <= 2 * i - 1; j++)
        {
            printf("* ");
        }
        printf("\n");
    }
}
/* Pattern 2: Inverted Pyramid */
void invertedPyramid()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for(i = n; i >= 1; i--)
    {
        for(j = 1; j <= n - i; j++)
        {
            printf(" ");
        }
        for(j = 1; j <= 2 * i - 1; j++)
        {

```

```

        printf("* ");
    }
    printf("\n");
}
}
/* Pattern 3: Diamond Pattern */
void diamond()
{
    int n, i, j;
    printf("Enter number of rows for upper half: ");
    scanf("%d", &n);
    /* Upper half */
    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n - i; j++)
        {
            printf(" ");
        }
        for(j = 1; j <= 2 * i - 1; j++)
        {
            printf("* ");
        }
        printf("\n");
    }
    /* Lower half */
    for(i = n - 1; i >= 1; i--)
    {
        for(j = 1; j <= n - i; j++)
        {
            printf(" ");
        }
        for(j = 1; j <= 2 * i - 1; j++)
        {
            printf("* ");
        }
        printf("\n");
    }
}
/* Pattern 4: Palindromic Number Pyramid */
void palindromicPyramid()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n - i; j++)
        {
            printf(" ");
        }
        for(j = 1; j <= i; j++)

```

```

        {
            printf("%d ", j);
        }
        for(j = i - 1; j >= 1; j--)
        {
            printf("%d ", j);
        }
        printf("\n");
    }
}

/* Pattern 5: Repeated-Number Pyramid */
void repeatedPyramid()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n - i; j++)
        {
            printf(" ");
        }
        for(j = 1; j <= 2 * i - 1; j++)
        {
            printf("%d ", i);
        }
        printf("\n");
    }
}

```

Task4: Output:

```
(kali㉿kali)-[~/hdoc27.c]
$ ./a.out

—PYRAMID/DIAMOND PATTERNS—
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit
Enter your choice: 1
Enter number of rows: 4
  *
 * * *
* * * * *
* * * * * * *

—PYRAMID/DIAMOND PATTERNS—
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit
Enter your choice: 2
Enter number of rows: 4
* * * * * * *
 * * * * *
  * * *
   *

—PYRAMID/DIAMOND PATTERNS—
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit
Enter your choice: 3
Enter number of rows for upper half: 4
  *
 * * *
* * * * *
* * * * * * *
 * * * * *
  * * *
   *
```

—PYRAMID/DIAMOND PATTERNS—

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit

Enter your choice: 4

Enter number of rows: 4

```
1
121
12321
1234321
```

—PYRAMID/DIAMOND PATTERNS—

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit

Enter your choice: 5

Enter number of rows: 4

```
1
222
33333
4444444
```

—PYRAMID/DIAMOND PATTERNS—

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit

Enter your choice: 1

Enter number of rows: 1

```
*
```

—PYRAMID/DIAMOND PATTERNS—

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit

Enter your choice: 3

Enter number of rows for upper half:

```
1
```

```
*
```

—PYRAMID/DIAMOND PATTERNS—

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit

Enter your choice: 4

Enter number of rows: 1

1

—PYRAMID/DIAMOND PATTERNS—

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit

Enter your choice: 5

Enter number of rows: 1

1

—PYRAMID/DIAMOND PATTERNS—

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit

Enter your choice: 7

Invalid choice!

—PYRAMID/DIAMOND PATTERNS—

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit

Enter your choice: 6

Exiting program...